

INTRODUCTION TO ANDROID PLATFORM



android



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- SmartPhone OS
- Most popular Brands
- What is Android
- Brief History
- Android Features
- Android Software Development
- The Developer Network

Google

ANDROID





CONTENTS

- Android Applications
- The Android Market
- Future possibilities
- Why Android is special?
- Other Possibilities

Google

ANDROID





SmartPhone OS

- ❖ Nokia's Symbian
- ❖ RIM's Blackberry OS
- ❖ Apple's iOS
- ❖ **Google's Android**
- ❖ HP's WebOS
- ❖ Microsoft's Windows Phone

symbian
OS

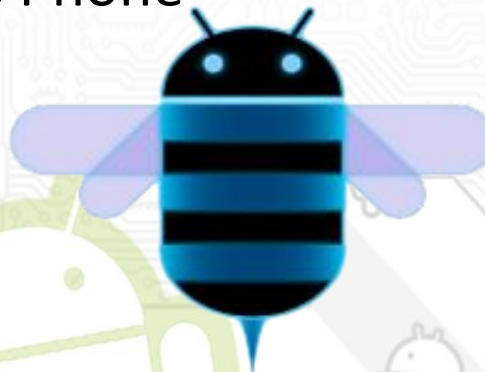
BlackBerry™

iOS 4

webOS™



Windows
phone



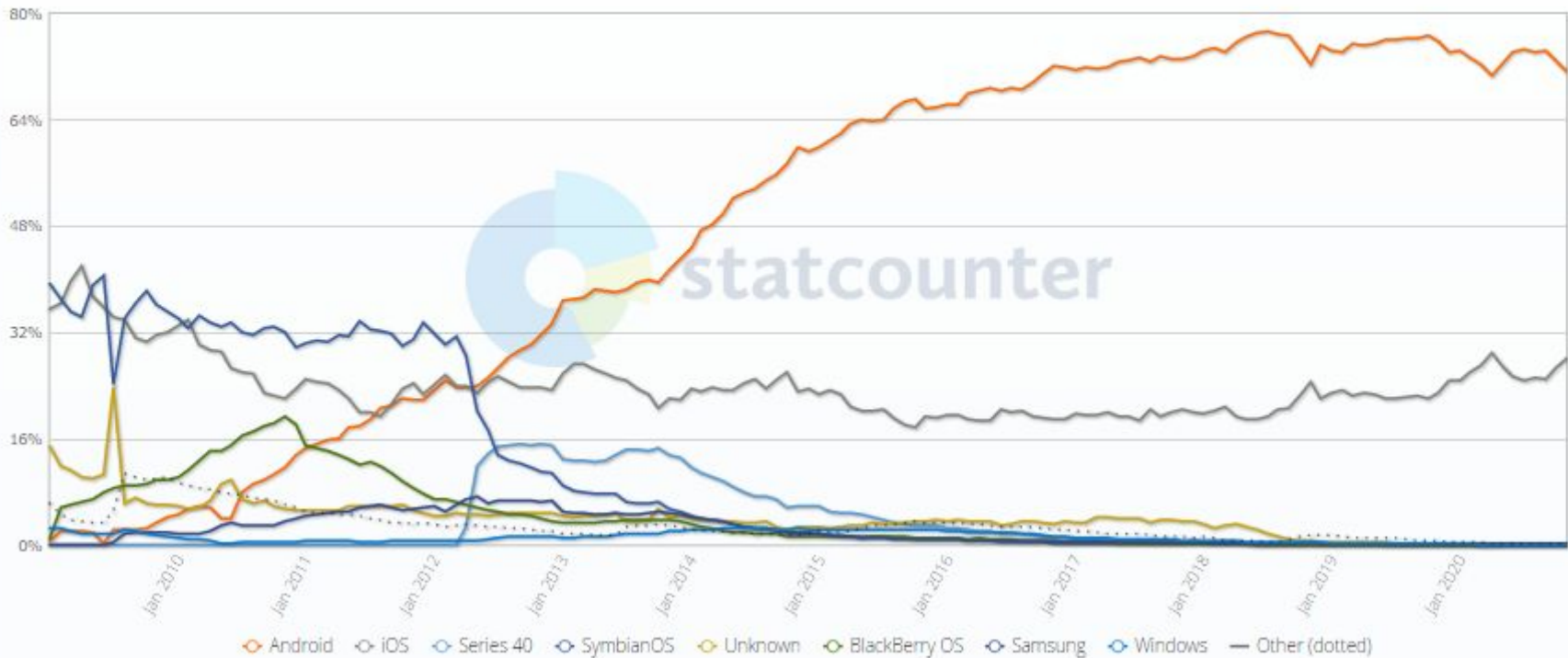


Android OS

Gooooo Android....

Mobile Operating System Market Share Worldwide

Jan 2009 - Nov 2020

[Edit Chart Data](#)

Android	iOS	Samsung	Unknown	KaiOS	Windows
72.92%	26.53%	0.22%	0.13%	0.07%	0.03%

Mobile Operating System Market Share Worldwide - October 2020



Most Popular Brands

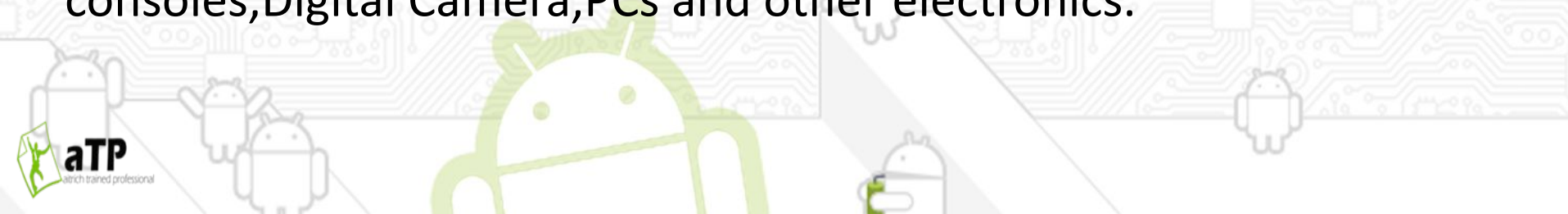
World's most popular smartphone brands, 2020 Q2

□ Huawei	20%
□ Samsung	20%
□ Apple	14%
□ Xiaomi	10%
□ Oppo	9%
□ vivo	8%
□ Lenovo	3%
□ Others	16%



Android OS

- ❖ Developer:-**Google**
- ❖ Android **Creators**:-Andy Rubin,Rich Miner,Nick Sears & Chris White.
- ❖ First phone launched **HTC G1** in **2008**.
- ❖ Average 1.5 million Android activations per day
- ❖ Over 1.5 billion smartphones were sold last year, with 1.3 billion of them being Android devices.
- ❖ 24000+ Different android devices.
- ❖ **Primarily** Touch screen Mobile devices(Smart phones,Tablets)
- ❖ **Now** Android Auto(for cars),Android TV,Wear OS(watches),game consoles,Digital Camera,PCs and other electronics.



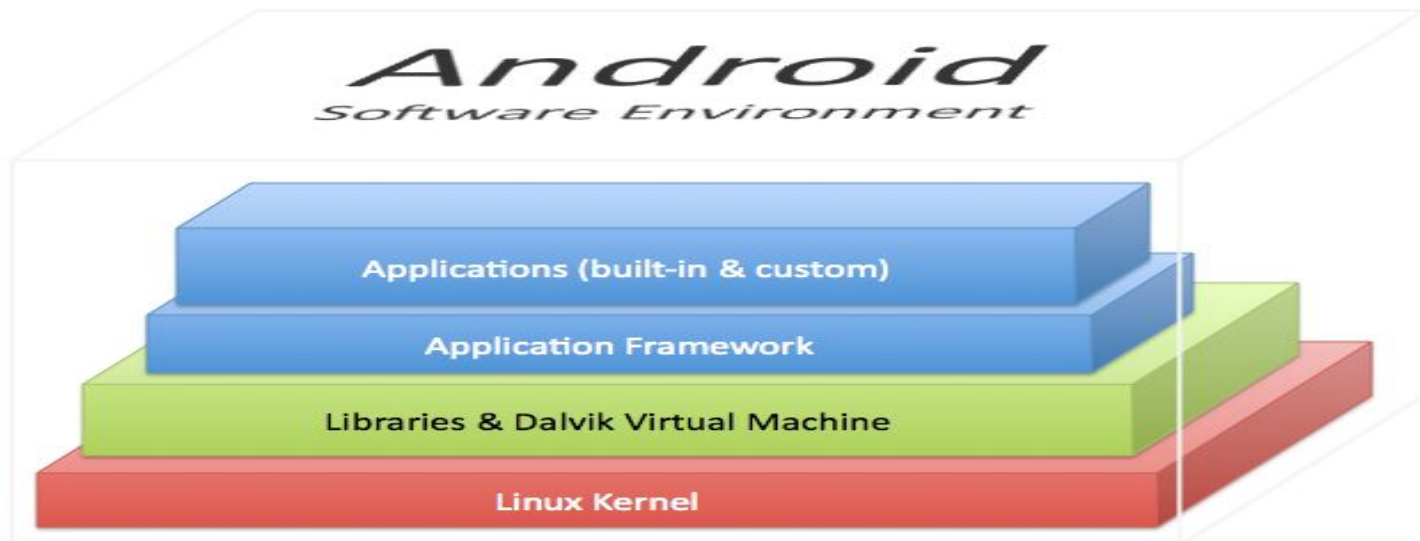


Why Android OS?

- Powerful Operating System.*
- Support Large number of Applications in SmartPhones.*
- Comfortable and advanced for the users.*
- Support more than 100 languages*
- Android is an open source system, which means that anyone,any mobile phone manufacturers can use and modify it.*
- Most Android applications have strong compatibility*

Development

- ❖ *Modified version of the **Linux kernel** and other open source software:-*
- ❖ Android relies on **Linux kernel** version 2.6.
- ❖ Built with Java, a well-supported and stable language with a wide base of existing developers.





Android SDK

- ❖ **Android Software Development Kit** used for developing User Applications
 - Provides the **tools and APIs**
 - Development Using **Java Programming language**.
 - **Kotlin** will likely be the leading language for Android within a few years.
- ❖ Linux provide core system services
 - Security
 - Power Management
 - Hardware Drivers

Linux™





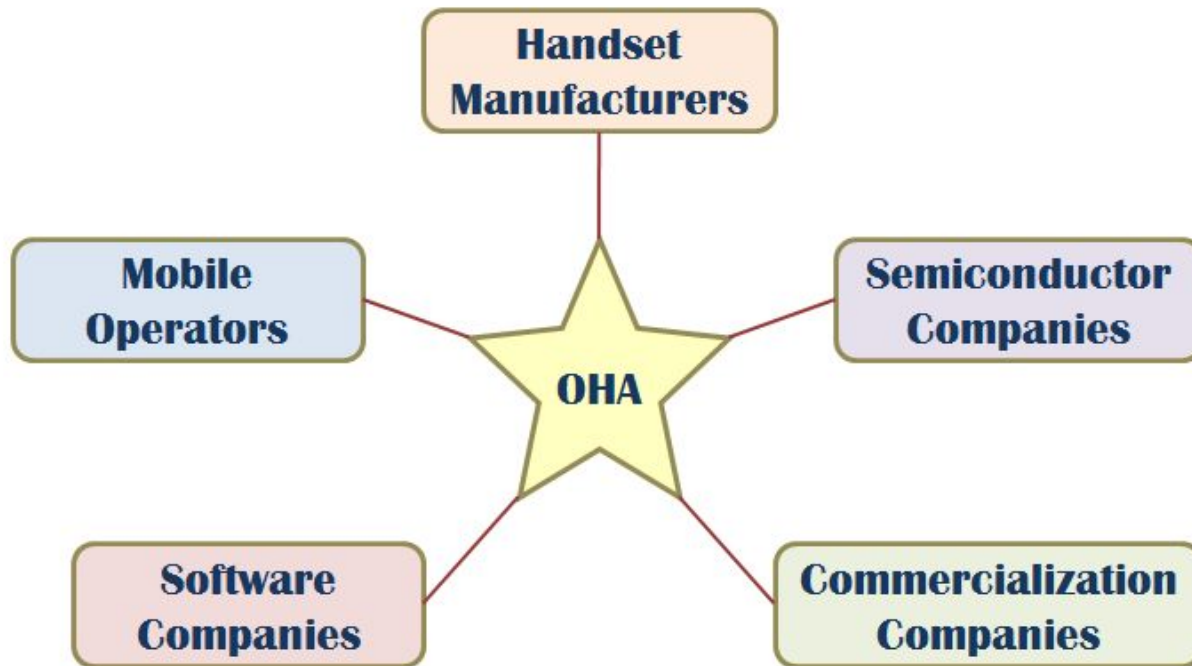
Open Handset Alliance



- An Alliance of **64** different hardware makers, carriers, and other mobile-related companies. They have come together to accelerate innovation in mobile and offer consumers a richer, less expensive, and better mobile experience.
- Est. 2007, led by Google
- Mission, Open Standards for Mobile Devices
- Headquarters - **South Korea**
- Region served - Worldwide



Open Handset Alliance



OHA...

Mobile Operators



Commercialization Companies



Handset Manufacturers



Software Companies



Semiconductor Companies





Brief History

◆ 2005

- Google acquired Android Inc. to start Android platform
- Work on Dalvik VM begins

◆ 2007

- Open Handset Alliance announced
- Early look at SDK

◆ 2008

- SDK 1.0 released(Astro)
- T-Mobile G1 announced (Android Phone)
- Android released open source (Apache License)
- Google sponsors 1st Android Developer Challenge

Brief History



❖ / 2009

❖ SDK 1.1(Bender) Rarely used.(**February**)



□ SDK 1.5 (**Cupcake**) - **April 27**

- New soft keyboard with “autocomplete” feature
- Third-party keyboard and Widgets were enabled and phone could upload directly to YouTube and Picasa.

□ SDK 1.6 (**Donut**) - **September 15**

- 
- It fixed reboot errors in the OS, revamped photo and video features (i.e. camera interface)
 - Featured better search integration.
 - Support for **larger screen sizes** and is the first version to offer Google's **turn-by-turn navigation** feature.
 - Support Wide VGA (Video Graphic Array for graphical display)

Brief History

- SDK 2.0(October)/2.0.1(December)

◆ 2010

□ SDK2.1 (Eclair) - January

- Include **Bluetooth 2.1**
- Support, **flash** and **digital zoom** for the **camera**, **multi-touch support**, **live wallpapers**, and more.

□ SDK2.2(Froyo)-May 20

- Improved speed by adopting the Javascript "**just-in-time**" compiler engine from Google's browser, Chrome.
- **GIF support** and Flash 10.1 plug-in support
- **USB tethering** and **Wi-Fi Hotspot** capability

□ SDK 2.3 (Gingerbread) - December 6

- **Nexus S**-Google co-developed with Samsung.
- SIP internet calling, NFC wireless transaction capability
- More than one camera, and gyroscopes and other sensors (barometers, gravimeters, and others are possible)

Brief History

◆ 2011

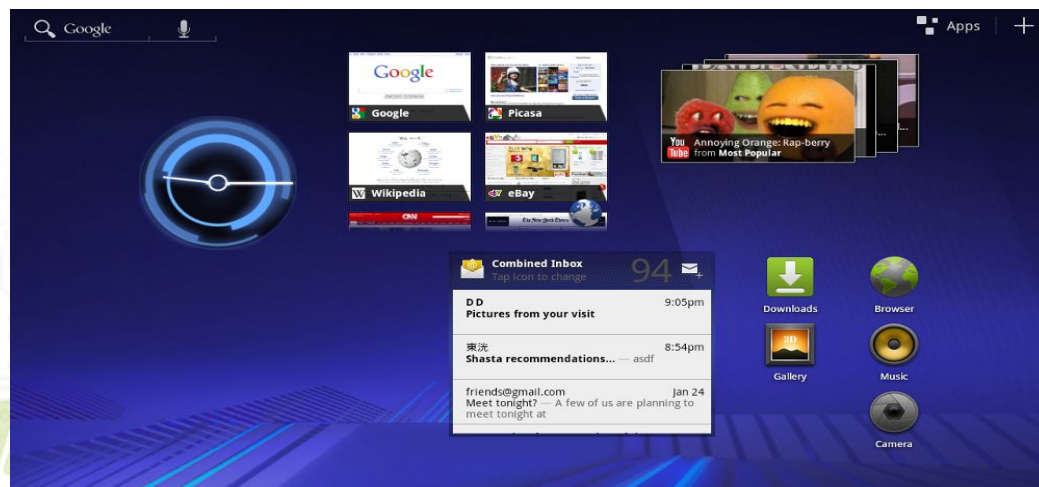
□ SDK 3.0 (Honeycomb) -February

• SDK 3.1(July),SDK3.2(August)

• It was made for tablets .

• **Motorola Xoom** was the first Android 3.X tablet to be released

• Support larger Screen, New UI for tablets, support multi-core processors





Brief History

❑ SDK 4.0 (Ice Cream Sandwich)-October 2011

- A **new design and default font**, as well as the ability to monitor and limit mobile data usage and other upgrades.



- Many devices were slow to adopt Ice Cream Sandwich
- 3 months after Ice Cream Sandwich, only one phone (Samsung Galaxy Nexus) had been released to run it.

◆ 2012

❑ SDK 4.1(Jelly Bean)- July 9

- Biggest changes included "Google Now," an AI assistant that anticipates your needs and better, more interactive notifications



- It allows "**voice typing**," a built-in speech-to-text engine that does not rely on Internet or data

Brief History

❖ 2013

□ SDK 4.4 - 4.4.4 (KitKat)-September 3-oct 31



- Ok Google allows people to access **Google Now** without even touching their phones, just verbally saying the phrase opens up the **artificial intelligence assistant**.
- Introduced **Emoji** to Google's keyboard.

❖ 2014

□ SDK 5.0 -5.1.1(Lollipop)-October 15



- A brand new runtime called **ART** that no longer relies on the older **DALVIK** runtime (which is somewhat based on Sun/Oracle specs). Lollipop also contains other UI improvements and has an excellent battery life on some devices
- UI improvements and has an excellent battery life on some devices.

Brief History

❖ 2015

- ❑ SDK 6.0(Marshmallow)-October 5
- ❑ Marshmallow 6.0.1-December 8



- Built-in support for fingerprint sensors, USB Type-C and power bank function. It also introduces Permissions control for apps, Doze power saver mode and lesser consumption of RAM.
- The addition of 200 emojis, quick camera launch, volume control enhancements
- Improvements to the UI of the tablet, and the correction made to the copy paste lag.

❖ 2016

- ❑ SDK 7.0(Nougat)-August 22,
- ❑ SDK7.1.2 (N2G48H) / December 5



- It is armed with a new Just-In-Time compiler based on the ART engine
- Unicode 9.0 Emoji support, and the new Vulkan 3D rendering API.

Brief History

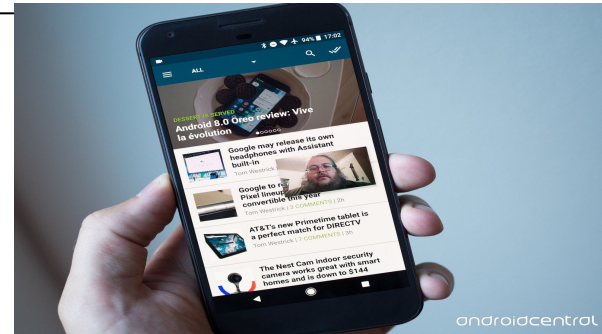
- Bundled notifications



2017

□ SDK 8.0(Oreo) -August 21

- Speed and efficiency.
- Picture-in-picture (PIP) mode(video playback)
- Multi-Window Lifecycle
- Managing notification channels.
- Autofill is a change that benefits both users and developers



2018

□ SDK 9.0(Pie)-March 7

- Adaptive battery life,App actions,App slices,
- New Gesture Navigation
- LockDown.



Brief History

❖ 2019

□ SDK 10.(android 10)- September 3

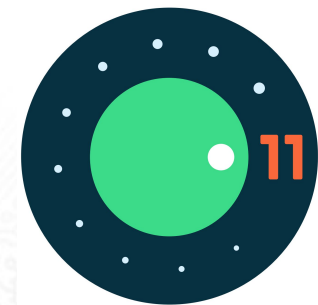
- Conversations are a priority.
- Built-in screen recording.
- Quick access to smart devices.
- Much better media controls.
- Permissions needn't be forever. ...
- Pixel only: Live view Location sharing. ...



❖ 2020

□ SDK 11(android 11)- September 8,,

- Smart Reply in all messaging apps.
- Enhanced location and privacy tools.
- Incognito mode for Google Maps.
- Live Caption.
- New parental controls. ...
- Edge-to-edge gestures.





SOFTWARE-FEATURES

Application framework



- ☐ The Application Framework layer provides many higher-level services to applications in the form of Java classes.
- ☐ Application developers are allowed to make use of these services in their applications.





Key-Services

- ❑ **Activity Manager** – Controls all aspects of the application lifecycle and activity stack.
- ❑ **Content Providers** – Allows applications to publish and share data with other applications.
- ❑ **Resource Manager** – Provides access to non-code embedded resources such as strings, color settings and user interface layouts.
- ❑ **Notifications Manager** – Allows applications to display alerts and notifications to the user.
- ❑ **View System** – An extensible set of views used to create application user interfaces.

Software-Features



◆ Integrated browser

- Based on the open source **WebKit engine**
- Ability to render pages in full (desktop) view
- Full CSS, JavaScript, DOM, AJAX support
- Support for single-column and adaptive view



rendering **Media support**

- A media framework based on PacketVideo OpenCore platform
- Support for standard video, audio, still-frame formats
- Support for hardware/software codec plug-ins





Software-Features

Dalvik virtual machine

- ❖ Discontinued Process Virtual Machine
- ❖ It is optimized for low memory requirements, and is designed to allow multiple VM instances to run at once.
- ❖ Its bytecode Format is still used as a distribution format, but no longer at runtime in newer android versions.
- ❖ **Dalvik VM** executes files in the Dalvik Executable (.dex) format





Android Run Time(ART)

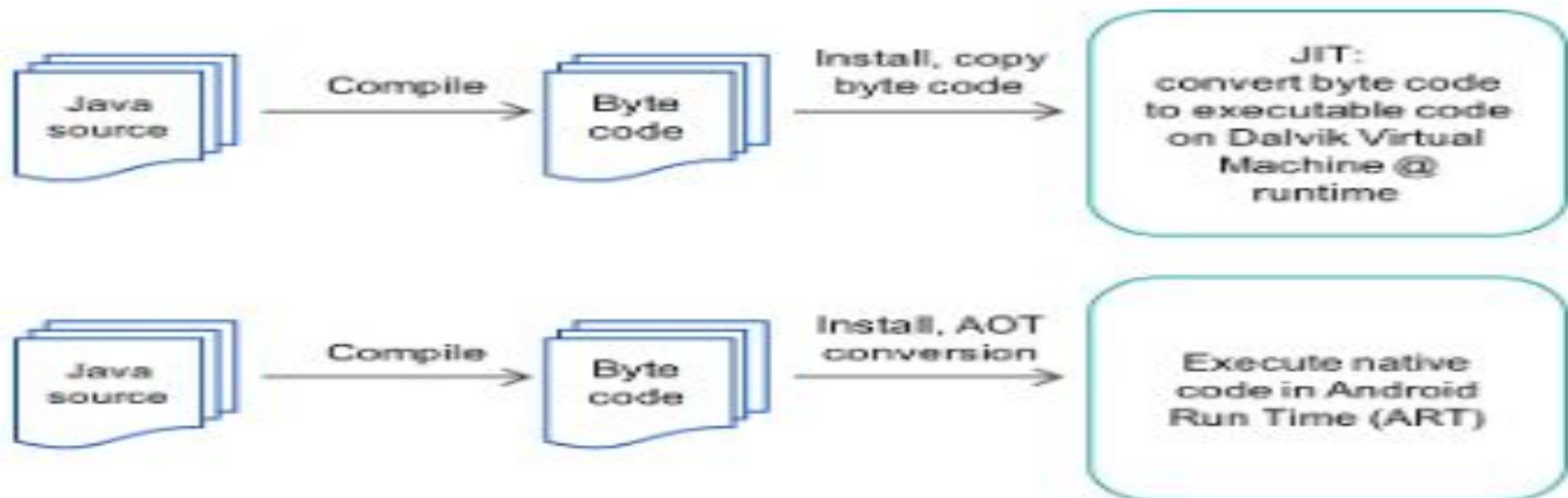
- ❖ ART is a new process virtual machine
- ❖ Introduced experimentally in the 4.4 (KitKat) release.
- ❖ It works based on Ahead-of-Time (AOT) compilation.
 - It is the act of compiling a Higher level Programming language in to a native (System-dependent) machine code.
- ❖ The Byte code is converted only once when the application is installed.
- ❖ This slows down the one-time activity of installing App, but gains the benefit of **faster execution at runtime**





Difference b/w Dalvik & ART

Dalvik virtual machine	ART
Based on Just-in-Time (JIT)	Based on Ahead-of-Time (AOT)
DEX bytecode translated into machine code during the app's execution	DEX bytecode has been translated into machine code during installation.
Need more battery charge	ART leads to a longer battery life
Need time for execution and launching	faster execution and shorter launch time
Less storage footprint	larger storage footprint



SQLite Database

SQLite



- It stores data to a text file on a device. Android comes in with built in SQLite database implementation.
- Don't need to establish any kind of connections for it like JDBC, ODBC e.t.c
- Powerful and lightweight (approx. 225kb) relational database engine (RDBMS).
- Its library is linked in an application program.
- It supports SQL syntax, transactions and prepared statements.





Hardware-Features

Cellular Networking :

◆ **GSM** (*G*lobal *S*ystem for *M*obile communications.)

- Cell phones use a cell phone service carriers GSM network by searching for cell phone towers in the nearby area.

◆ **1G**

- Uses **FDMA**(Frequency Division Multiple Access)technique.
- Data speed 2.4kbps(Analog signal transmission)
- Poor voice quality & security



Cellular Networking

◆ 2G

- Uses **DMA**(Digital Multiple Access)technique:-**CDMA**(Code Division Multiple Access)/**TDMA**(Time Division Multiple Access)
- Introduced **SMS**,Internet Browsing.
- Data Speed **50kbps**.

◆ 3G

- **WCDMA**(Wideband Code Division Multiple Access)technique, increase in **BandWidth**.
- DataSpeed **2mbps**.



Cellular Networking

- It was the Huge step in the transformation of Basic phone to a **SmartPhone**.
- It allows the transfer of data for uses Such as,
 - GPS,Videos,Voice Call...

◆ 4G

- **OFDMA**(Orthogonal Frequency Division Multiple Access).
- MIMO(Multiple Input Multiple Output)technology:-

Multiple antennas are used at both the source (transmitter) and the destination (receiver).
- DataSpeed 20-100mbps,suitable for high resolution movie and T.V



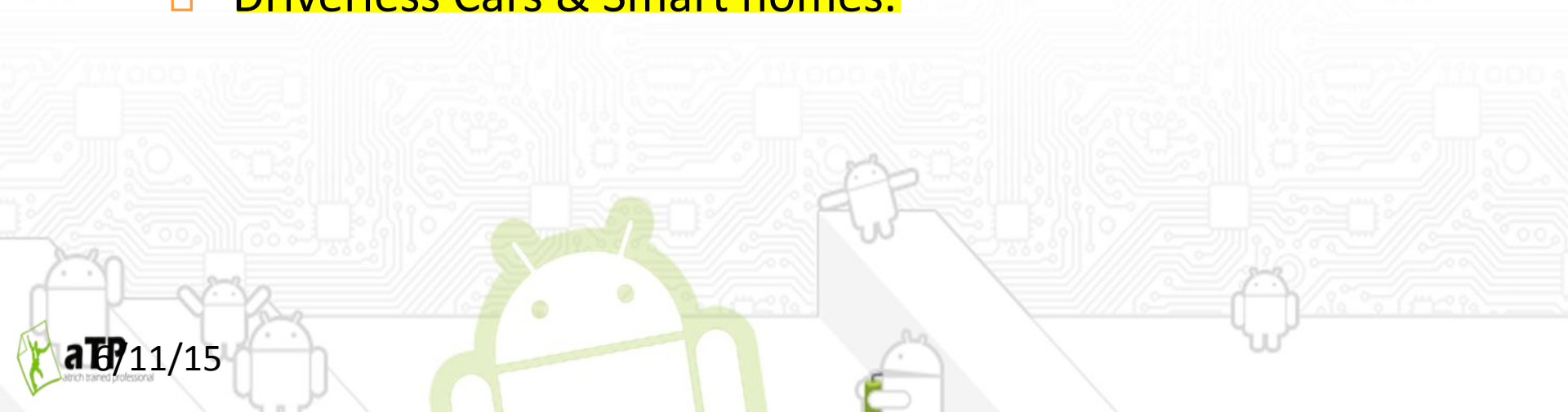
Cellular Networking

❖ 5G

□ Rolling out soon

- Use Enhanced MIMO Technology & Millimeter Waves
- It will provide Seamless Connectivity to support the Internet of things.

□ Driverless Cars & Smart homes.



LAN

Bluetooth:- Open wireless technology standard for exchanging data over short distances from fixed and mobile devices.

- Create personal area networks with high levels of security.

❖ **Wi-Fi:-** A wireless standard for connecting electronic devices.

- A Wi-Fi enabled device such as a personal computer, video game console, Smartphone, and digital audio player can connect to the Internet when within range of a wireless network connected to the Internet.

❖ **NFC:-** Provides access to **Near Field Communication** (NFC) functionality, allowing applications to read NDEF (NFC Data Exchange Format) message in NFC tags. A "tag" may actually be another device that appears as a tag.

HardWare Facilities

- ❖ Graphics Hardware Acceleration
 - Is a techniques used to improve performance
- ❖ Camera
- ❖ GPS - The **global positioning system** is a satellite-based navigation system
- ❖ Compass - A **compass** is a navigational instrument for determining direction relative to the Earth's magnetic poles.
- ❖ Touch Screen and Accelerometer for motion sensing





Android Security

- ❖ Android is multi-process system, in which each application runs in its own process
- ❖ Most security at the process level through standard Linux facilities (Linux user ID and group ID) that are assigned to applications.

Additional securities through “*permission*”

- Mechanism that enforces restrictions on the specific operations that a particular process can perform.



Android SDK

- ❖ A Software development kit that enables developers to create applications for the Android platform.
- ❖ It provides the **tools** and **APIs** needed to begin developing applications
- ❖ The Android SDK includes :
 - Android Emulator
 - Development tools
 - Documentation
 - Sample Applications





Android SDK

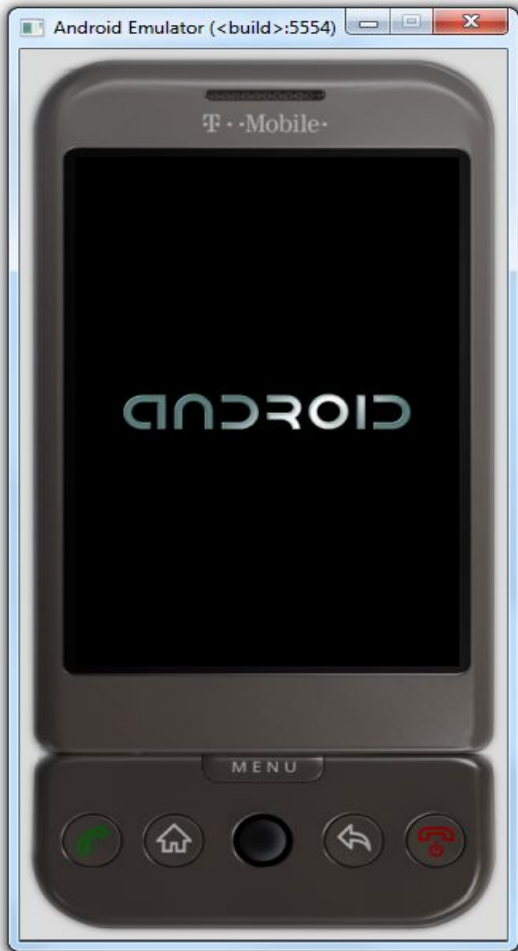
Android Emulator or Android Virtual Device (AVD)

- ❖ A virtual mobile device that runs on your computer.
- ❖ You can develop and test Android applications without using a physical device.
- ❖ Android SDK and AVD Manager allows you to create AVDs that target any Android API level.



Android SDK

Android Emulator



- ❖ QEMU-based ARM emulator
- ❖ Interact with the emulated mobile device just as you would an actual mobile device, using Mouse and Keyboard

Limitations

- No Camera support
- No Support for Bluetooth

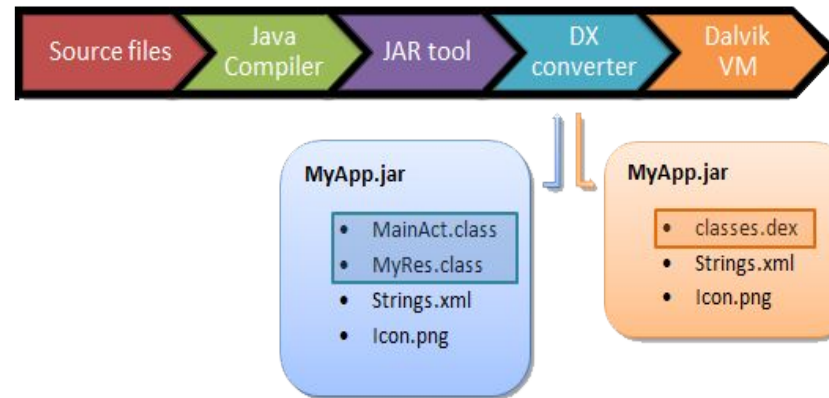


Android SDK

Development Tools

❖ Dalvik Cross Assembler (**dx**)

- Used for merging and converting Java-Standard-bytecode Classes (*.class) into one single binary file (*.dex) that can be executed by the Dalvik VM.



❖ Android Asset Packaging Tool (**aapt**)

- Packages Android project files into .apk files for installation on the emulator and handset.

Android SDK

Development Tools

❖ Android Debug Bridge(**adb**)

- A client-server tool that interacts directly with Android devices and emulators using a command-line interface

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Ilhan>cd c:\android\tools

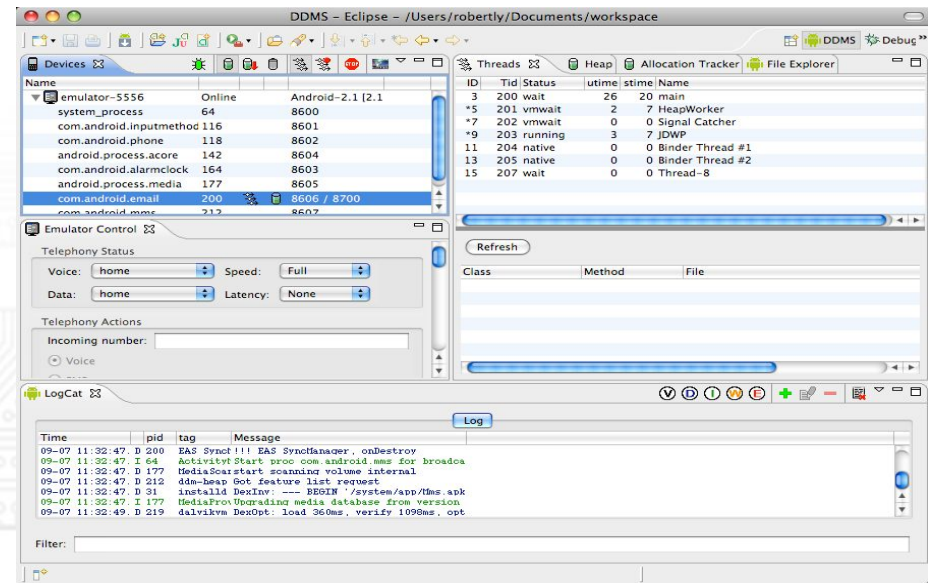
C:\android\tools>adb devices
* daemon not running. starting it now *
* daemon started successfully *
List of devices attached
5700c85e73a5    device

C:\android\tools>adb install touiteur.apk
674 KB/s (816115 bytes in 1.181s)
pkg: /data/local/tmp/touiteur.apk
Success

C:\android\tools>
```

❖ Dalvik Debug Monitor Server (**ddms**)

- Used to Debug Android Application

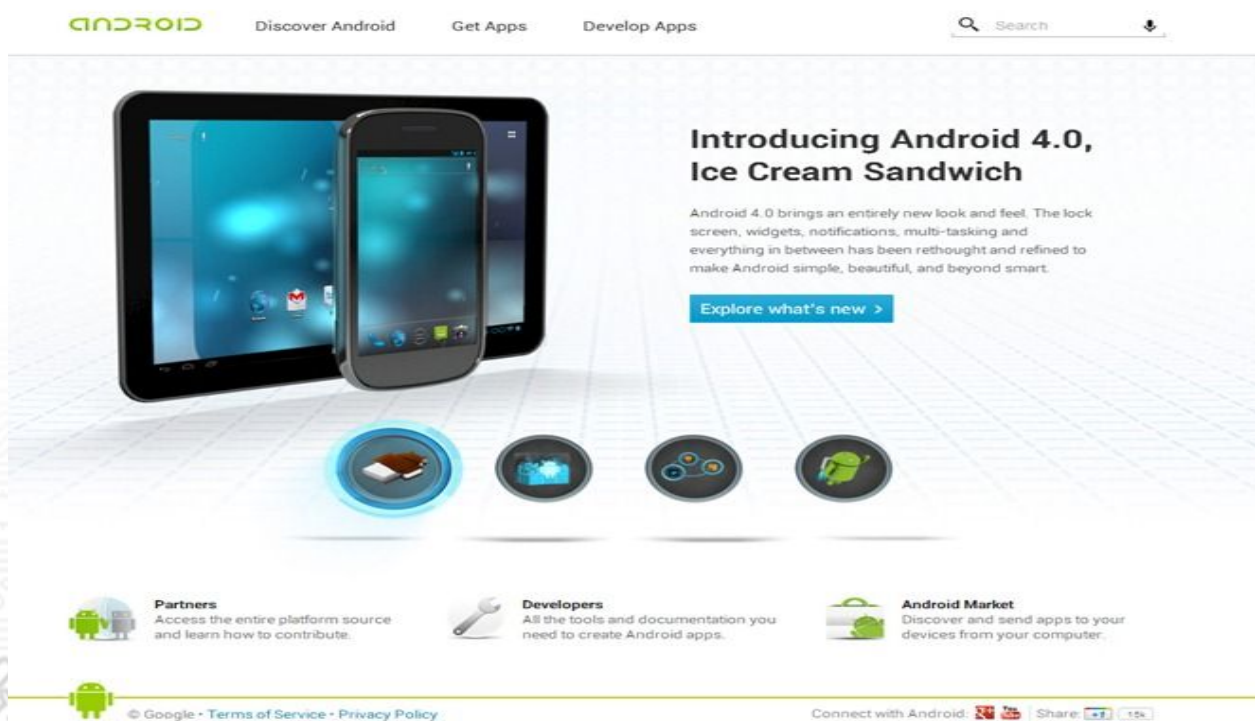




Android SDK

Documentation

- ❖ The Android documentation is provided in HTML format locally and online at <http://developer.android.com>





Android SDK



Documentation - Divided into 7 main Sections:

- ❖ **Home** – Provide developer announcements and important links
- ❖ **SDK** – provides information about the different Android SDK versions available, Android Native Development Kit (NDK)
- ❖ **Dev Guide** – This links to the Android Developers Guide
- ❖ **Reference** – provides a listing of all the Android APIs with detailed coverage of specific classes and interfaces.
- ❖ **Resources** – provides access to Android technical articles and tutorials
- ❖ **Videos** – which is available online only, is your resource for Android training videos.
- ❖ **Blogs** – published by the Android development team.



Software Development

Development Tools

- ❖ Android Studio
- ❖ Android SDK



Programming Languages

- ❖ **Kotlin** – Officially Supported
- ❖ C/C++/C#/Scala(Scalable Language)- also possible
- ❖ Java





Software development

Android Studio

- Android Studio is a new Android development environment based on IntelliJ IDEA.
- It provides new features and improvements over Eclipse ADT



Android
Studio



Android Developer Network

- ❖ There are countless Android Communities online, which help for Android development.
- ❖ These are most helpful for finding
 - Additional Resources
 - Interacting with other developers of various experience levels
 - Finding solutions to common problems.
 - For getting help with specific problems.
- ❖ If you have any Android development questions, they have probably been answered in these Android Groups

Android Developer Network



Some places to find help with developing your android application

- ❖ Official developer site-<http://developer.android.com/index.html>
- ❖ IRC – #android-dev on freenode

As detailed in the resource tab

<http://developer.android.com/resources/community-more.html>,

- ❖ Google Groups - <http://groups.google.com/group/android-developers>
- ❖ Stack Overflow - <http://stackoverflow.com/questions/tagged/android>
- ❖ Twitter - <http://twitter.com/androiddev>



Distributing Android app



- ❖ Google Play Store
- ❖ Self-publish on your own website
- ❖ Use a third party application store





Android Market

The Official Store: Google Android Market

https://play.google.com/store/apps/details?id=com.google.android.finsky&hl=en_IN

Gives you an easy way to distribute your applications to Android users. Key features of Android Market include:

Easy and simple to use
Start using Android Market in 3 easy steps: register, upload, and publish.

Come one, Come all
Android Market is open to all Android application development. Once registered, developers have complete control over when and how they make their applications available to users.

Greater visibility
Developers can manage their applications portfolio where they can view information about downloads, ratings and comments. Developers can publish updates and versions of their apps.



Android Market

- ❖ \$25 registration fee for developers
- ❖ Developer receives 70% of each sale
- ❖ Remaining amount goes to carriers
- ❖ Google does not take a percentage



Deliver apps quickly;
Saves time & money

Real-time benefits;
Broader audience

Platform Independent framework;
Maximize the reach

Improves productivity many times even for a single smartphone OS

Google Play Store



https://play.google.com/store/apps?hl=en



Apps

Categories ▾

Home

Top Charts

New Releases



My apps

Shop

Games

Family

Editors' Choice

Account

Payment methods

My subscriptions

Redeem

Buy gift card

My wishlist

My Play activity

Parent Guide

Sports video games

See more



Carrom Pool
Miniclip.com

★★★★★



Pooking - Billiards
MOUNTAIN GAME

★★★★★



Football Strike - Multiplayer
Miniclip.com

★★★★★



FIFA Soccer
ELECTRONIC ARTS

★★★★★

Car games

See more



Traffic Racer



Asphalt Xtreme: Rally

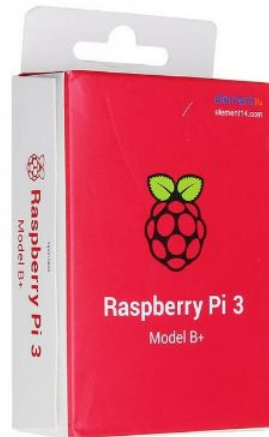


Race Max



Racing in Car 2

Android Devices





Future Possibilities

- ❖ Google Android Revenue is going to Overtake iPhone in the coming years.
- ❖ The OHA is committed to make their vision a reality
 - That is to deploy the Android platform for every mobile operator, handset manufacturers and developers to build innovative devices.
- ❖ Intel doesn't want to lose ownership of the netbook market, so they will be preparing for anything, including Android.

What Makes Android Special?

Open



Industry

Developer

Industry

- Software stack open-sourced under Apache 2.0 license
- Source available after first handsets ship
- Anyone will be able to build a system image

Users

- Users have control of their experience
- They control what gets installed
- They choose the defaults

Developer

- Don not need permission to ship an application
- No hidden or privileged framework APIs
- Can integrate, extend and replace existing components



Resources

Learn Android

- ❖ <http://developer.android.com/guide/basics/what-is-android.html>
- ❖ http://en.androidwiki.com/wiki/Main_Page
- ❖ <http://www.openhandsetalliance.com/>

